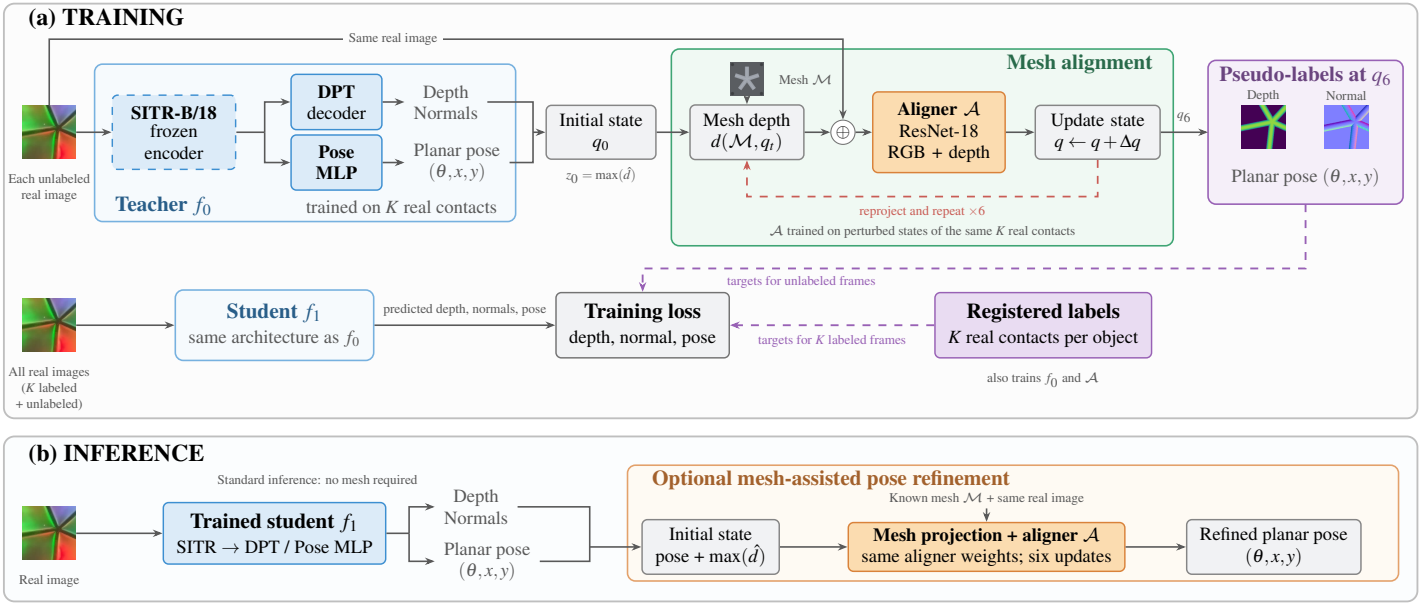

(a) TRAINING
Same real image
Each unlabeled real image
SITR-B/18 frozen encoder
DPT decoder
Pose MLP
Depth
Normals
Planar pose (θ, x, y)
Teacher f_0
trained on K real contacts
Initial state q_0
$z_0 = \max(\hat{d})$
Mesh alignment
Mesh $\mathcal{M}$
Mesh depth $d(\mathcal{M}, q_t)$
Aligner $\mathcal{A}$ ResNet-18 RGB + depth
Update state $q \leftarrow q + \Delta q$
reproject and repeat ×6
$\mathcal{A}$ trained on perturbed states of the same K real contacts
q_6
Pseudo-labels at q_6
Depth
Normal
Planar pose (θ, x, y)
targets for unlabeled frames
All real images (K labeled + unlabeled)
Student f_1
same architecture as f_0
predicted depth, normals, pose
Training loss
depth, normal, pose
targets for K labeled frames
Registered labels
K real contacts per object
also trains f_0 and $\mathcal{A}$
(b) INFERENCE
Standard inference: no mesh required
Real image
Trained student f_1
SITR → DPT / Pose MLP
Depth
Normals
Planar pose (θ, x, y)
Optional mesh-assisted pose refinement
Known mesh $\mathcal{M}$ + same real image
Initial state pose + $\max(\hat{d})$
Mesh projection + aligner $\mathcal{A}$
same aligner weights; six updates
Refined planar pose (θ, x, y)